Dealing with Space in Multi–Agent Systems: a model for Situated MAS

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ABSTRACT

The paper introduces the Multilayered Multi–Agent Situated System (MMASS). The MMASS allows tha description of situated agents that is, agents sensitive to the spatial relationships that determine constraints and abilities for actions as well as privileged cooperation relationships. The main feature of the MMASS model is to give an explicit definition of the spatial structure of the environment in which the system of situated agents acts and interacts. Agent environment, possibly multilayered, can reproduce a physical space. It is nevertheless possible to define a 'virtual space' that agents can roam and where they interact. Interactions take place when two or more agents are brought into dynamic spatial relationship through a set of reciprocal actions, and have an influence on the future behavior of the agents. There are several application domains that require the representation of the space in order to take advantage in the multi–agent approach. Typical examples of these domains are the Multi–Agent Based Simulation (MABS) and the Computer Supported Cooperative Work (CSCW). Two specific problems of these domains will be used in the paper to illustrate the MMASS model.

1. INTRODUCTION

The main aim of this work is to present the *Multilayered Multi–Agent Situated Systems* (MMASS) model. A situated agent lives in an environment where the space is explicitly structured. In [11] the concept of situated agent is introduced, without any explicit reference to the spatial structure of the environment. At our knowledge no other proposal in the literature dealing with interaction between agents and their environment or modelling the environment takes into account its spatial structure. We claim that there are several application domains that require an explicit representation of the spatial structure of the environment in order to take advantage of a multi–agent approach.

The MMASS model describes the spatial structure of the environment as a multilayered network of sites. Moreover, it defines a set of influences generated by agents and propagating along the edges of the network and leaving on sites information about their presence. Agents situated in this environment are strongly influenced by their position that is, the site of the space in which they are situated. This kind of Multi–Agent System (MAS) is based on a perception–deliberation–action mechanism, that describes agents as characterized by a set of possible actions, and on a mechanism for the selection of the action to be undertaken based on the internal state and the position of the agents themselves. In particular, the position in the environment defines the situation in which agents are acting, where "situation" refers to a potentially complex combination of internal and external events and states [6]. Situated MAS are generally made of reactive agents that is, elementary memoryless agents with a defined position in time and space. Reactive situated agents perform their actions as a consequence of the perception of signals coming either from other agents or from the environment, and are sensitive to the spatial relationships that determine constraints and abilities for actions as well as privileged cooperation relationships. The environment where agents are situated can reproduce a physical space. It is nevertheless possible to define a "virtual space" that agents can roam and where they interact. Interactions take place when two or more agents are brought into dynamic spatial relationship through a set of reciprocal actions, and have an influence on the future behavior of the agents.

A formalized description of the basic principles of the MMASS model together with examples of its application are reported in this paper. This work is part of a larger project whose aim is to develop a tool to support the design, development and execution of applications according to the MMASS approach. The work presented here gives the basis to provide designers with a language and a framework for the development of systems of agents that are characterized and influenced by their spatial position and where the spatial relationship among agents is considered in agent interaction.

Typical examples of application domains in which the structure of the environment must be considered are Multi–Agent Based Simulation (MABS) and Computer Supported Cooperative Work (CSCW) [5].

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MABS is an agent approach to simulation. It is supported by the idea that it is possible to represent the global behavior of a dynamical system as the emergent property resulting from the interaction of its constituents acting locally, i.e. as the result of the interaction of an assembly of autonomous agents with their own operational autonomy. MAS have been used to simulate artificial worlds [9] as well as natural or social phenomena [10, 15]. Within the MABS framework, the application of the MMASS model to the simulation of the localization process of shopping centers will be shown. Finding a "good" location for a shopping center (a well known problem that has been widely studied by urbanists and economists and recently tackled by different computational approaches [7]) means to consider properties of the space, such as the presence of roads, the population density distribution on urban areas, the distance from competitors. Shopping centers are modelled as competitive agents and the influence of their potential location is given in terms of the gravitational model [8]: agents emit gravitational fields that attract consumers or repel competitors. A previous model of this problem by Cellular Automata (CA) [18] has been experienced, but the MMASS approach here presented allows to by–pass some intrinsic restrictions required by the CA approach (for more detail on the comparison between CA and MMASS see [4]).

In the CSCW framework, empirical studies [14] have highlighted how people use forms of coordination based on their perception (the so called awareness) of the presence and distance of the other people both in the physical space and in the 'logical space' identified by the considered application. Examples of 'logical space' are the partial order of activities of a workflow, the organization chart establishing the relationships among roles, the abstract representation of a physical layout where resources are located, and so on. Here the distance plays a fundamental role in identifying a suitable technological solution supporting this recently uncovered form of cooperation, as distance is one of the main parameters that can be used to tune and filter the amount of awareness information to be sent or perceived by an entity occupying a specific site in the logical space. The CSCW literature proposes models of coordination based on a space (called Spatial Models of Awareness since [12]). However, these proposals focus on solutions to specific problems and give a limited formalization of the underlying model [16]. The approach proposed here provides designers with primitives that allow the construction of the functionalities devoted to promote awareness in cooperative applications, thus overcoming the above limits. An example will illustrate this claim.

2. THE MULTILAYERED MULTI–AGENT SITUATED SYSTEM

A *Multilayered Multi–Agent Situated System (MMASS)* can be defined as a constellation of interacting *Situated MAS* (MASS) through the primitive:

$$\mathbf{Construct}(MASS_1 \dots MASS_n)$$

Thus, a MMASS is defined by the set of its constituting *Situated MAS*'s ($\langle MASS_1 \dots MASS_n \rangle$).

A MMASS is a system of agents whose main features are:

- situated agents: each agent is associated with a site of the MASS space and the agent behavior is strongly influenced by its position in the topological structure of this environment;
- agent interactions within the MMASS are spatial dependent: agent behavior is influenced by other agents (i.e. by their presence or by the signals they emit), and both type of interactions are strongly dependent of the spatial structure of the agent environment. Agent presence is perceived only in the agent neighborhood while signals propagate according the topological structure of the environment;
- agent behavior: both agent state and position can be changed by the agent itself according to a perception–deliberation–action mechanism. Thus, each reactive agent constituting the MMASS after the perception of signals emitted by other agents, selects the local action to be undertaken (according to its state, position and type) and executes it;
- heterogeneity: agents are also characterized by a type, that determines their perceptive sensitivity and capability. Thus, a MASS is an heterogeneous system of agents, where agents with different features and capabilities coexist and interact. Moreover, agent environment can be characterized by a multilayered topological structure that is composed by multiple, heterogeneous and interconnected topological spaces.

A MASS of a Multilayered MASS is defined by the triple

$$< Space, F, A >$$

where *Space* models the environment where the set A of agents is situated, acts autonomously and interacts via the propagation of the set F of fields. Thus a MMASS can be also denoted by:

$$\langle < Space_1, F_1, A_1 > \dots < Space_n, F_n, A_n > \rangle$$

where,

- $n = |Space|$ and $\left(\bigcup_{i \in Space} Space_i\right)$ is the set of *Spaces* defining the multilayered structure of the MMASS;
- $\left(\bigcup_{i \in F} F_i\right)$ denotes the set of *fields* acting in the MMASS;
- $\left(\bigcup_{i \in A} A_i\right)$ denotes the set of *agents* situated in it.

Field emission–propagation–perception is the mechanism defined for asynchronous interaction among agents situated in the same or different MASS's: an agent emits a field that propagates throughout the *Space* and can be perceived by other agents. In order to allow MASS interaction the MMASS model introduces the notion of *interface*. The interface of a MASS specifies fields imported from and exported into each MASS, and takes the following format:

$$\mathbf{Interface}(MASS_i, \mathbf{export} : \mathbf{E}; \mathbf{import} : \mathbf{I})$$

where $E \subset F_i$ and $I \subset F_i$ are respectively the set of fields exported into and imported from $MASS_i$. Imported fields are used in agent actions as internal fields are. As will be better explained in section 2.4, they can be mentioned in agent actions and used in the definition of agent perception

function (see section 2.3). By definition, the value of an external field in any site of the local *Space* of a MASS is the value specified at its emission. Moreover, the receiving MASS has to define wether and how this value has to be internally propagated by means of local fields defined for this purpose. In fact, their distribution function (see section 2.2) is highly dependent on the structure of the local *Space* which is completely hidden to external MASS's.

In sections 2.1, 2.2, 2.3 each component of a MASS will be described in more detail.

2.1 Spaces

The environment where the agents of a MASS are situated is named *Space* and is defined as made up of a set P of sites arranged in a network. Each *site* $p \in P$ can contain at most one agent and is defined by the 3–tuple

$$< a_p, F_p, P_p >$$

where:

- $a_p \in A \cup \{\perp\}$ is the agent situated in p ($a_p = \perp$ when no agent is situated in p that is, p is empty);
- $F_p \subset F$ is the set of fields active in p ($F_p = \emptyset$ when no field is active in p);
- $P_p \subset P$ is the set of sites adjacent to p.

In this way the *Space* can be considered as a not oriented graph of sites.

2.2 Fields

Fields acting within a MASS can be generated by agents of the MASS, can have a source outside the local *Space* in the case of fields imported from another MASS or from outside the MMASS in the case of open systems. Each field of a MASS is characterized by the set of values that the field can assume during its propagation throughout the *Space*. Propagation occurs according to the diffusion function that characterizes the field and that specifies how its values propagate throughout the space according to its spatial structure. Moreover, field comparison and field composition functions are defined in order to allow field manipulation.

A field $f \in F$ is defined by the 4–tuple

$$< W_f, Diffusion_f, Compare_f, Compose_f >$$

where:

- W_f denotes the set of values that the field can assume;
- $Diffusion_f : P \times W_f \times P \rightarrow (W_f)^+$ is the diffusion function of the field computing the value of a field on a given site taking into account in which site and with which value it has been generated. Since the structure of a *Space* is generally not regular and paths of different length can connect each pair of sites, $Diffusion_f$ returns a number of values depending on the number of paths connecting the source site with each other site. Hence, each site can receive different values of the same field along different paths.
- $Compare_f : W_f \times W_f \rightarrow \{True, False\}$ is the function that compares field values. For instance, in order to verify whether an agent can perceive a field value (see the definition of agent perception in the following subsection) the value of a field at a site and agent sensitivity threshold are compared.
- $Compose_f : (W_f)^+ \rightarrow W_f$ expresses how field values have to be combined (for instance, in order to obtain the unique value of the field at a site).

2.3 Agents

The *Space* of each MASS is populated by a set A of individuals called *agents*. An agent $a \in A$ is defined by the 3–tuple

$$< s, p, \tau >$$

where:

- τ is the *agent type*;
- $s \in \Sigma_\tau$ denotes the *agent state* and can assume one of the values specified by its type;
- $p \in P$ is the site of the *Space* where the agent is situated.

Agent *type* describes the set of states the agent can assume, a function to express agent sensitivity to fields, and the set of actions that the agent can perform. Thus, an agent type τ is defined by the 3–tuple

$$< \Sigma_\tau, Perception_\tau, Action_\tau >$$

where:

- Σ_τ defines the set of states that agents of type τ can assume;
- $Perception_\tau : \Sigma_\tau \rightarrow [\mathbf{N} \times W_{f_1}] \ldots [\mathbf{N} \times W_{f_{|F|}}]$ is a function associating to each agent state a vector of pairs in which, for the i–th pair, the first element expresses a coefficient to be applied to the field value (f_i), and the second one expresses the agent sensibility threshold to f_i in the given state (let $c^i_\tau(s)$ and $t^i_\tau(s)$ be their names). This means that an agent of type τ in state $s \in \Sigma_\tau$ can perceive a field f_i only when it is verified

  $$Compare_{f_i}(c^i_\tau(s) \cdot w_{f_i}, t^i_\tau(s))$$

  that is, when the first component of the i–th pair of the perception function ($c^i_\tau(s)$) multiplied for the received field value w_{f_i} is greater than the second component of the pair ($t^i_\tau(s)$).
- $Actions_\tau$ denotes the set of actions that agents of type τ can perform (see action description in the following subsection).

2.4 Actions

$Actions_\tau$ specifies whether and how agents change their state and/or position, how they interact with other agents, and how neighboring agents can influence them. In general, actions have two possible purposes: they can be undertaken by an agent in order to modify its state or position (*intra–agent actions*), or to interact with other agents in both

synchronous or asynchronous ways (*inter–agent actions*). Specifically, *trigger* and *transport* are the intra–agent actions, and in particular *trigger* defines how the perception of a field causes a change of state in the receiving agent, while *transport* defines how the perception of a field causes a change of position in the receiving agent. *emit* and *react* are the inter–agent actions. The first one is asynchronous and at–a–distance while the second one is synchronous and can occur only among neighboring agents.

As previously introduced, *asynchronous interaction* among agents takes place through a field emission–propagation–perception mechanism. An agent emits a field that is, it generates a field defining its parameters (intensity values, diffusion function, and so on), when its state is such that it can be source for it. This constitutes one side of the asynchronous interaction among agents: that is, the sending of broadcast messages by an agent. Field values propagate throughout the space according to the diffusion function of the field. Field diffusion along the space allows the other agents to perceive it. $Perception_\tau$ function, characterizing each agent type, defines the second side of an asynchronous interaction among agents: that is, the possible reception of broadcast messages conveyed through a field, if the sensitivity of the agent to the field is such that it can perceive it. This means that a field can be neglected by an agent of type τ if its value at the site where the agent is situated is less than the sensitivity threshold computed by the second component of the $Perception_\tau$ function. This is the very essence of the broadcast interaction pattern, in which messages are not addressed to specific receivers but potentially to all agents populating the space.

Reaction defines the *synchronous interaction* among a set of agents characterized by given states and types and pair–wise situated in adjacent sites (that is, *adjacent agents*). Synchronous interaction is a two–steps process. Reaction among a set of agents takes place through the execution of a protocol introduced in order to synchronize the set of autonomous agents. The synchronization protocol involves a set of adjacent agents. When an agent wants to react with the set of its adjacent agents since their types satisfy some required condition, it starts an *agreement* process whose output is the subset of its adjacent agents that have agreed to react. An agent agreement occurs when the agent is not involved in other actions or reactions and when its state is such that this specific reaction could take place. The agreement process is followed by the synchronous reaction of the set of agents that have agreed to it.

In order to define agent action set ($Actions_\tau$) let us consider an agent $a =< s, p, \tau >$ that is, an agent of type τ whose current position is site p and whose current state is s. Moreover, $p =< a, F_p, P_p >$ where F_p is the set of fields active in p and P_p is the set of sites adjacent to p. To define the four action above outlined, we will use operators of the form *action–condit–effect*. *action* is an expression of the form $f(x_1 \dots x_n)$ where f specifies the action name and x_i are variables which can appear in *condit* and *effect* expressions. *condit* and *effect* express respectively the set of conditions that must be verified in order to let the agent execute the action, and the set of effects deriving from the execution of the action. They are sets of atomic formula $p(a_1 \dots a_k)$ where p is a predicate of arity k and a_i are either constants or variables. According to this syntax, the four basic agent actions are:

1. $action : trigger(s, f_i, s')$
   $condit : state(s), perceive(f_i)$
   $effect : state(s')$

   where $state(s)$ and $perceive(f_i)$ are verified when the agent state is s, and $f_i \in F_p$ and $Compare_{f_i}\left(c^i_\tau(s) \cdot w_{f_i}, t^i_\tau(s)\right) = True$ (that is, the field f_i is active in p ($f_i \in F_p$) and agents of type τ in state s can perceive it). The effect of a trigger action is a change in state of the agent according to the third parameter.

2. $action : transport(p, f_i, q)$
   $condit : position(p), empty(q), near(p, q), perceive(f_i)$
   $effect : position(q), empty(p)$

   where $perceive(f_i)$ has the same meaning as in *trigger*, while $position(p)$, $empty(q)$ and $near(p, q)$ are verified when the agent position is p, $q \in P_p$ and $q =< \perp, F_q, P_q >$ (q is a site adjacent to p and no agent is situated in it). The effect of the execution of a transport action is the change in position of the agent undertaking the action and, as a consequence, the change of the local space where the agent is situated. In fact let us suppose that agent a has executed a $transport(p, f_i, p')$. This means that a has changed its position from p to p' and as a consequences sites p and p' have changed to

   $p =< \perp, F_p, P_p >$
   $p' =< a, F_{p'}, P_{p'} >$

3. $action : emit(s, f, p)$
   $condit : state(s), position(p)$
   $effect : added(f, p)$

   where $state(s)$ and $position(p)$ are verified when the agent state and position are s and p.

   The effect of a emit action ($added(f, p)$) is a change at each site of the space according to $Diffusion_f$. In particular the set of fields in p where the emitting agent is situated changes to

   $$p =< a, F_p \diamond f, P_p >$$

   where the operator $F \diamond f$ simply adds a field f to a field set F if the field does not already belong to the set. Otherwise it composes its already present field value(s) and the new one. Let w_f^{old} and w_f^{new} be their names,
   $F \diamond f = F \cup f$ and

   $w_f = w_f^{new}$ if $f \notin F$ (before the execution of the emit action) while
   $w_f = Compose(w_f^{old}, w_f^{new})$ if $f \in F$.

4. $action : reaction(s, a_{p_1}, a_{p_2}, \dots, a_{p_n}, s')$
   $condit : state(s), agreed(a_{p_1}, a_{p_2}, \dots, a_{p_n})$
   $effect : state(s')$

where $state(s)$ and $agreed(a_{p_1}, a_{p_2}, \ldots, a_{p_n})$ are verified when the agent state is s and agents situated in sites $\{p_1, p_2, \ldots, p_n\} \subset P_p$ have previously agreed to undertake a synchronous reaction.

The effect of a reaction is the synchronous change in state of the involved agents; in particular, agent a changes its state s'.

2.5 Adding features to the basic MMASS

The basic MMASS model presented above has been extended to incorporate relevant features of the target systems. Typically, the application of the MMASS model to specific problems and domains required to introduce the possibility to dynamically modify both the spatial structure of the environment and the number of agents constituting each MASS. To this aim agent types can be characterized by the following additional actions:

- *AddSite(p)*: the agent adds a site p to its neighborhood, i.e., the set of sites adjacent to the site in which it is situated. Since agents have only a local visibility of their environment (i.e. they ignore the existence of sites not adjacent to the one hosting them) this action can be performed only after that the agent has obtained by other agents the needed information about a spatially not adjacent site (typically through a reaction or the perception of a field).
- *DeleteSite(p)*: the agent deletes a site p from its neighborhood (obviously, this action can be performed only when the site is empty that is, it is not occupied by any agent).
- *Clone(a)*: the agent creates an agent, specifying its type (only agents of the same type can be created), its initial state (one of the states specifiable for its type), and its position (one of the site of the neighborhood of the creating agent);
- *Clear(self)*: the agent deletes itself. This action has been introduced for sake of garbage collecting agents that are no more operative. This fact is denoted by a specific state (i.e. 'dead'). The agent executing this action simply clears itself and makes empty its site.

3. MMASS APPLICATIONS

3.1 Simulation

The MMASS model has been applied in the MABS framework to the problem of the location of suitable sites for extra–urban shopping centers [2]. Intuitively, a suitable location for a shopping center should be accessible through a comfortable road system, close enough to large residential areas, far enough from other shopping centers and large enough to host the mall and related facilities and services (e.g. parking lots, restaurants, gas stations). Simulations can be useful when suitable space is available and a good location for a new shopping center has to be chosen, or when the possibility of success in locating a new shopping center in an area already served by other retailers has to be evaluated [7]. Geographical factors, such as the proximity of other centers and residential areas, are essential for the choice of a suitable location for a shopping center. Moreover, once some centers have settled in a given area, other factors like competition should be taken into account.

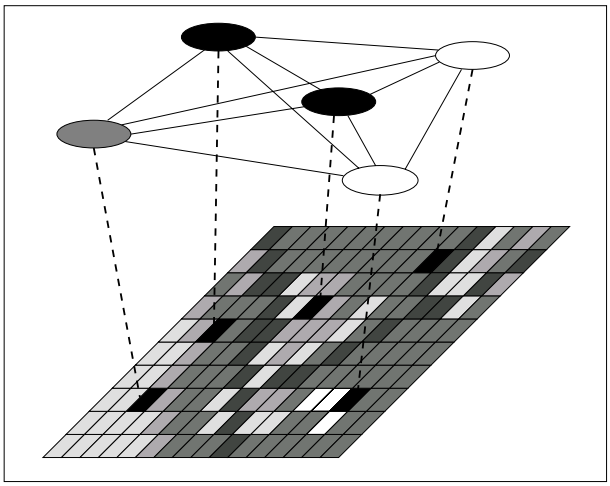


Figure 1: The Multilayered MASS composed by the *Territorial MASS* and the *Strategic MASS*. Shopping centers are represented in both MASS's: each black site of the Territorial MASS is connected by its strategic counterpart in the graph of the Strategic MASS.

In the following it will be shown how the design of a MMASS composed by two MASS's allows the simulation of the two aspects involved in this problem. Two MASS's have been defined: the *Territorial MASS* and the *Strategic MASS* (see Figure 1 for a representation of the two layers of the resulting Multilayered MASS). In the Territorial MASS, the formation or the disappearance of a shopping center is modelled by considering only geographical factors, while in the Strategic MASS already existing centers compete with one another trying to attract consumers. Each shopping center is represented in both the Territorial and the Strategic MASS. Interaction between the two MASS's is performed through field emission by agents belonging to a MASS and their perception as imported fields by agents of the other MASS.

3.1.1 The Territorial MASS

The Territorial MASS represents the territory where the simulation is performed. It is defined as a regular grid that is, the territory is divided into square blocks, each one populated by an agent. Then for each site $p =< a_p, F_p, P_p >$, $a_p \neq \perp$ and $\|P_p\| = 4$.

Agent types reproduce the different types of area that can be found in an urban environment, that is: *Residential*, *Industrial*, *NotBuildable* (e.g. mountains or rivers or parks), *Road* and *Suitable* (that is areas suitable for a shopping center installation). The first four types have only one possible state, which does not change during simulations. On the contrary, agents of type *Suitable* model the formation (or the disappearance) of shopping centers by changing their state that can assume one of three possible values: *Empty*, *Small* and *Large*.

Agents of each type emit fields f_{Sm} and f_{Lg} according to their type and state in order to attract and repel agents of type *Suitable*. Positive field values of f_{Sm} (f_{Lg}) favor the formation and growth of a small (large) shopping center while

f_{Sm}	$Distance1$	$Distance2$	$Distance3$
Residential	9	9	1
Industrial	1	2	0
Protected	−6	2	−2
Road	6	−1	−10
Large	−9	−7	−1
Small	−8	−4	2
Empty	3	−8	−8

Table 1: Values of field f_{Sm} according to the type of the emitting agent and the distance from the source site. The value of the field at distance greater than three sites is zero.

negative values have a negative influence. Field values and their distribution functions change according to the type of the emitting agent. Possible values of f_{Sm}, computed according to [8], are shown in Table 1. For instance, the presence of a large shopping center has a negative impact on the location of other centers in its surroundings, while residential areas strongly encourage the formation of shopping centers with positive field values. Every field is perceived only by *Suitable* agents that, as a consequence, change their state. For instance, if an agent in state *Empty* perceives a field value higher than a given threshold, it changes its state and becomes a *Small* or a *Large* shopping center, depending on which field has been perceived.

In order to let the Territorial MASS interact with the Strategic MASS, fields f_{Ter} and f_{Str} have been defined. *Suitable* agents of the Territorial MASS emit a f_{Ter} field when some changes in its state must be notified to its counterpart belonging to the Strategic MASS. This means that no agent of the Territorial MASS can perceive f_{Ter} fields but these fields are exported to the Strategic MASS. On the contrary, *Suitable* agents representing small or large shopping centers can perceive f_{Str} fields that are emitted in the Strategic MASS and imported into the Territorial MASS when some change occurs as effect of shopping centers competition (see the following subsection for more details). In this way, the MMASS composed by the Territorial and the Strategic MASS's is provided with a mechanism for the two MASS interaction without the loss of autonomy of the two composing MASS's.

Some of the actions contained in $Action_{Suitable}$ set are:
$emit(Empty, < 3, Diffusion_{f_{Sm}}, +, <>, p)$
$trigger(Empty, f_{Sm}, Small)$
$trigger(Empty, f_{Lg}, Large)$

This means that *Suitable* agents in *Empty* state emit fields f_{Sm} with source value 3 (to positively influence the formation of small shopping centers). Moreover they change their state from *Empty* to *Small* as a consequence of the perception of a positive value of the same field while they change to *Large* when they perceive a positive value of field f_{Lg}.

3.1.2 The Strategic MASS

Geographical factors, such as the proximity of other centers and residential areas, are essential for the choice of a suitable location, but once some centers have settled in a given area other factors, like competition, should be taken into account. For example, one center could start low price special sales in order to attract consumers, or invest heavily on advertisement. Thus, the MMASS model allows to take into account the competition among centers already established in a given territory through the definition of the *Strategic* layer of the MMASS.

The number of sites of the Strategic MASS is equal to the number of *Suitable* agents of the Territorial MASS. The space of this MASS is complete that is, each pair of sites is connected by an edge. Moreover, in each site is situated a *Strategic* agent and it is connected to the corresponding site of the Territorial MASS in which is situated a *Suitable* agent. *Strategic* agent states are denoted by the couple $(x, f(x))$ where x is an integer value expressing the power of the agent and $f(x)$ is a function of the agent power and can assume one of the three values *Inactive*, *Defensive*, and *Aggressive*.

The state of each *Strategic* agent corresponding to an empty area on the Territorial MASS is $(0, Inactive)$. At the beginning of the simulation, when no shopping center is settled, all agents are in this state. When an *Inactive* agent perceives a f_{Ter} field, it is activated and changes its state either to *Aggressive* or *Defensive* with power proportional to the intensity of field f_{Ter} perceived.

Now, competition is modelled as a reaction between agents situated in neighboring sites. When at least one agent is *Aggressive*, it tries to "attack" the other shopping centers. The rule we adopted is simple: the most powerful one wins, and increases its power by a given value set by the user, while the loser has its power decreased by the same value.

When some change in Strategic agent state occurs, agents of this MASS emit a field (f_{Str}) with intensity corresponding to their new power value that is perceived by its counterpart on the Territorial MASS. When the field value perceived by the shopping center on the Territorial MASS drops below a given threshold, the center disappears.

3.2 Computer Supported Cooperative Work

This section illustrates how the features of the MMASS model can be used to construct mechanisms to support the promotion of awareness [13, 16] within and across cooperative applications [17]. These mechanisms take facts from the applications and elaborate them in order to produce awareness information; the latter is passed back to the applications and is presented at the user interface. Since in general, cooperating actors are involved in several applications at the same time, the multilayered structure of the MMASS is used to represent the awareness spaces associated to them to produce awareness information. Each space is populated by the proxies of entities pertaining to the associated application. Their behavior is expressed as actions of the MMASS model. Facts are interpreted as imported fields and the involved entities are always sensitive to them.

In order illustrate how the above mechanisms can be constructed we consider a workflow application (WFMS) whose awareness space is defined in a natural way as a logical space populated by actors, roles and tasks. The topology of the space is defined by the relations expressing causal dependencies between tasks, assignment of responsibility for tasks to roles and allocation of actors to tasks. Moreover, tasks to be executed concurrently are also connected for sake of awareness promotion. In the same organization it is likely that another application supports the collaborative management of human resources (HRM). For sake of conciseness, we do not specify its internal structure. These two applications have associated an awareness space as shown in Figure 2.

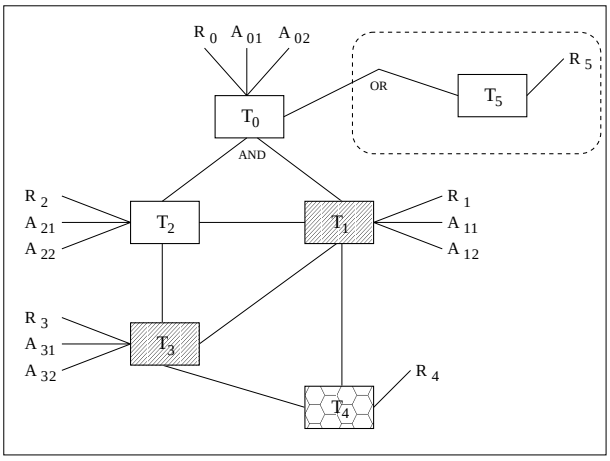


Figure 2: The awareness space associated to the workflow after the handling of Fact(α), Fact(β) concerning T_1 and T_3, and facts managing the conclusion of T_0 and T_2

Let us consider them in turn and sketch how a whole (quite complex) mechanism can be constructed to promote awareness among the actors involved in the workflow and a specific role located in the second space. In the awareness space associated to the WFMS, each type τ has the following sets of states
$\Sigma_{Task} = \{enacted, future, concluded, in_trouble\}$
$\Sigma_{Actor} = \{doing, done, available\}$
$\Sigma_{Role} = \{assigned_to, in_charge, done\}$

In the following we give some examples of the $Actions_\tau$ set and of the $Perception_\tau$ function characterizing the above types.
The enactment and execution of the WFMS generate facts that are elaborated by the model in order to compute which awareness information has to reach each agent on the basis of its state, sensitivity and position in the (logical) awareness space. This means that the standard external field associated to each incoming fact activates a sequence of actions (script) to elaborate it. Examples of such fact–script pairs are considered here below.

$Fact(\alpha)$ = the selection of a branch of an OR node. For example in Figure 2 the left branch is selected. The rule associated to this event activates a space manipulation primitive (i.e. *DeleteSite*) disconnecting the right branch of the workflow. This portion of space becomes unreachable from now on for any awareness purpose.

$Fact(\beta)$ = task T_i is enacted by R_i allocating $A_{i1} \dots A_{ik}$ to T_i. This event is interpreted by the following two actions:
1. Each involved actor A_{ij} executes:
$transport(p_j, f, q)$
where the perception of the field f corresponds to receiving the fact notification from the workflow application; q belongs to the neighborhood of the site where T_i is located.
2. Then A_{ij}, T_i and R_i execute respectively the following reactions which result in synchronous change of their states:
$reaction(available, A_{i1}, .., A_{i(j-1)}, A_{i(j+1)}, .., A_{ik}, R_i, T_i, doing)$
$reaction(future, A_{i1}, \dots, A_{ik}, R_i, enacted)$
$reaction(assigned_to(T_i), A_{i1}, \dots, A_{ik}, T_i, in_charge(T_i))$
The fact concerning task conclusion can be treated analogously by the execution of a reaction by A_{ij}, T_i and R_i, leading them to the state *done*, *concluded* and *done* respectively.

$Fact(\gamma)$ = Task T_i is going beyond its deadline, its state becomes critical. The awareness information generated by this event is as follow: the task changes its state and propagates a warning information to the other agents, with strength depending on how much they are involved that is, logically distant. This event is elaborated by T_i according to the following script:
1.$trigger(enacted, f, in_trouble)$
where the perception of the field f is the reception of the fact from the application.
2.$emit(in_trouble, TR, p_i)$
where TR assumes values in **N** (the set of integers); *Compose* is the sum and *Compare* is the less than relation in **N**; and

$$Diffusion_{TR}(s_0, 2, s) = \begin{cases} 2 & if\ s = s_0 \\ \frac{2}{dist(s,s_0)} & otherwise \end{cases}$$

where s_0 is the site where the field has been emitted (e.g. p_i where T_i is situated in the above case), and $dist(s, s_0)$ is the distance between s_0 and any site s. TR values are made decrease with the distance since the bigger the distance between entities the weaker the logical relation linking them.

So for example, if T_1 is *in_trouble* then it emits the field TR with 2 as initial value. The value at the sites where R_1, A_{11} and A_{12} are situated is 2, while it is 1 at the sites where R_j and A_{j1} and A_{j2} (with $j \neq 1$) are located.

As an example of the perception function, let us consider the type *Role* and its sensitivity to the above field TR:

$$Perception^{TR}_{Role}(done) = (0, 1)$$

$$Perception^{TR}_{Role}(assigned_to) = (1, 1)$$

$$Perception^{TR}_{Role}(in_charge_of) = (2, 3)$$

The above definition of the sensitivity functions 'modulates' the perception of the field TR by the *Role* agents in the various states: in fact, they can cancel, maintain or amplify its values by playing as a coefficient for the values of TR at the sites where agents are situated.

Suppose that the information conveyed by the above TR field has to reach the awareness space associated to the HRM application. The awareness space of the HRM application contains a role, called *Personnel Manager*, which is sensitive to TR in all its possible states, with different thresholds associated to them. When it is situated in any site p and perceives TR it changes its state to *trouble_handling* and decides to assign an additional actor which is in the state *available* to the task T_1 that is *in_trouble*. Then, it emits a field $Assign_to(task)$, which is exported to the awareness space of WFMS. The following script belongs to its set of Actions:
1 $trigger(any_state, TR, trouble_handling)$
2 $emit(trouble_handling, Assign_to(T1), p)$

This implies that the type *Actor* in the WFMS awareness space has to be sensitive to $Assign_to(task)$ when it is in the

state *available* and its set of Actions contains the following script:
$trigger(available, Assign_to(T1), doing)$
$transport(p, doing, q)$
where q is an empty site and belongs to the neighborhood of the site where T_1 is located.

Finally, the combination of the two awareness spaces is obtained by the **Construct**() and **Interface**() primitives as follows:
$Construct(WFMS, HRM)$
$Interface(WFMS,$
$export : TR, Import : Assign_to(task))$
$Interface(HRM,$
$export : Assign_to(task); import : TR)$

4. CONCLUSIONS AND FUTURE WORK

A model for systems of situated agents (i.e. MMASS), has been presented. Agent behavior occurring, for instance, as response to perception of fields generated by other agents and propagating throughout the environment, is strongly influenced by the spatial structure of the latter. Since field values can decrease during field propagation, agents perceive them depending on the site of the space where the agent is situated and the site in which the field has been generated. Moreover, an explicit definition of the spatial structure of agent environment allows the definition of distance and adjacency among situated agents. The MMASS model provides two types of interaction among agents constituting the MAS (i.e. reaction and field emission–propagation–perception), and both types take into account the spatial relationship between agents. In fact, fields influence agents depending on their spatial distance, and reactions occur only when agents are spatially adjacent. Finally, since the environment is in some case a shared resource for agents, its spatial structure constraints also in other agent actions, like agent change of position in the environment.

Future work will concerns the development of a language and a tool to design, develop and execute applications according to the MMASS approach. The support tool is currently under design. The preliminary language specification and a discussion on mechanisms to address the issues to develop a framework that implements the model can be found in [3] and [1] respectively.

Moreover, the MMASS approach application will contemporaneously proceed within the MABS and CSCW domains. In particular, it will be followed to model and simulate complex system dynamics in the theoretical immunology domain (project funded by MURST, the Italian Ministry for University and Scientific Research) and for socio–economic analysis (result of a collaboration with the Austrian Research Center Seibersdorf).

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